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Toward an AI-assisted Comprehensive Meta-analysis of the Organizational Commitment Literature

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With special thanks to Behrad Khorramnazari (The Synthesis Company of California) for his work on this project.

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Agenda

Where we've been, where we are, and where this is going

- Brief History of Commitment Theory & Research
- Challenges to conducting comprehensive reviews
- Toward an AI-Assisted Meta-Analysis
  - Where we are
  - Where we are going
  - What is possible

Organizational commitment: the early years

A brief history — some milestones

1960s–70s	Multidisciplinary interest in commitment; multiple competing conceptualizations
1979	Mowday, Steers & Porter introduce the OCQ — the seminal book follows in 1982
1990	Mathieu & Zajac publish the first comprehensive meta-analysis
1990–91	Allen & Meyer introduce the TCM measures (1990) and the three-component model (1991)
2002	Meyer, Stanley, Herscovitch & Topolnytsky — comprehensive meta-analysis of TCM research

Two comprehensive meta-analyses

Studies **(k)** and participants **(N)** per analysis — Mathieu & Zajac (1990) and Meyer et al. (2002)

1990

Mathieu & Zajac (1990)

STUDIES PER ANALYSIS (k)

3–43

PARTICIPANTS PER ANALYSIS (N)

302–15,531

VARIABLES META-ANALYZED **48**

26 | 14 | 8

2002

Meyer et al. (2002)

STUDIES PER ANALYSIS (k)

3–69

PARTICIPANTS PER ANALYSIS (N)

441–23,656

VARIABLES META-ANALYZED **38**

19 | 10 | 9

antecedents · correlates · consequences

Mathieu & Zajac (1990), Psychological Bulletin, 108(2) · Meyer et al. (2002), Journal of Vocational Behavior, 61

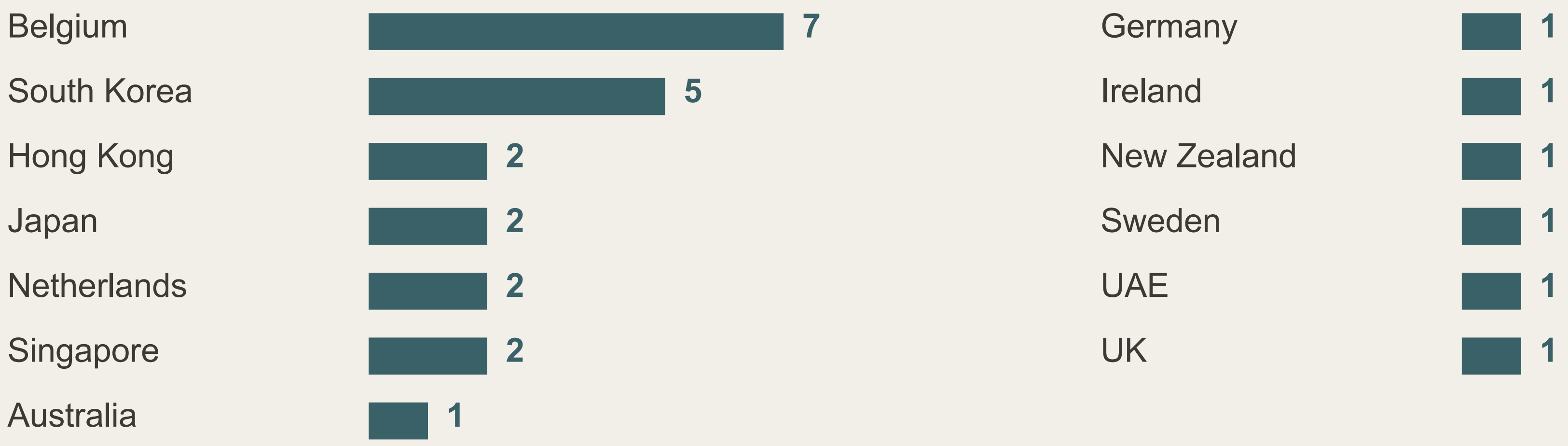
A modest cross-national base

Meyer et al. (2002) — studies reporting country of origin

GEOGRAPHIC REACH OF THE EVIDENCE

13 countries · **27** studies

STUDIES BY COUNTRY (k)



Studies reporting country of origin (n = 27); tabulation from the ICAP 2014 presentation.

Challenges to conducting comprehensive reviews

Why a truly comprehensive commitment review has become impractical by hand.

1

Studies keep growing

Commitment research piles up faster every decade.

2

Reviews turn narrow

Meta-analyses cover fewer variables to stay feasible.

3

Even narrow is too much

Now even targeted reviews outstrip human effort.

4

AI shares the labour

AI screens and codes at scale, clearing the bottleneck.

An AI-assisted pipeline — with humans in the loop

Screening and coding were performed by Eva, a family of AI agents; the team set the rules and decided every uncertain case.

1	→	2	→	3	→	4	→	5
Search		**De-duplicate**		**Screen**		**Code**		**Validate**
7 databases, keyword + citation		consolidate overlapping hits		every record; flags its own uncertainty		82 structured fields per abstract		automated consistency checks

HUMANS STAY IN THE LOOP

- Set & refine the inclusion rules
- Decide every case the AI flags as uncertain
- Pilot-review the coding before each full run

The AI does the volume; the team sets the rules and owns the judgment calls.

FINDING THE STUDIES

Literature Search

A Global Search

Literature search method

Two complementary search strategies in every database (except ERIC)

- **Keyword & subject-term search —** truncated organizational terms (org*, work*, employ*, job*, firm*, institution*, affect*, norm*, continu*) within three words of commit*, plus each database's subject headings; syntax adapted to each platform.
- **Cited-reference search —** every record citing any of 14 foundational commitment works (e.g., Porter et al., 1974; Mowday et al., 1979; Allen & Meyer, 1990; Meyer & Allen, 1991; Meyer et al., 2002; Klein et al., 2014).
- **Scholarly sources only —** academic journals, books, dissertations & theses, conference papers, reports, and working papers.
- **All languages —** the search applied no language restriction, so abstracts in any language were captured, not only English.

Seven databases

Searched December 20, 2025 – January 7, 2026

ABI/INFORM Collection	ProQuest
Business Source Complete	EBSCO
ERIC	Ovid · keyword only
Dissertations & Theses	ProQuest
APA PsycINFO	Ovid
Scopus	Elsevier
Web of Science	Clarivate

Records identified, by database and search strategy

Records retrieved per database, before consolidation and de-duplication · bars share one scale

Database	Keyword search	Cited-reference search
ABI/INFORM Collection	26,659	9,642
Business Source Complete	29,551	6,382
ERIC	5,372	— not supported
Dissertations & Theses	17,936	20,410
APA PsycINFO	16,257	5,516
Scopus	25,611	19,505
Web of Science	23,898	15,660
Total	**145,284**	**77,115**

Combined: **222,399 records**

ERIC (Ovid) does not support cited-reference searching.

From 222,399 records to 22,484 coded abstracts

De-duplication only removes overlap — AI relevance screening did the real narrowing

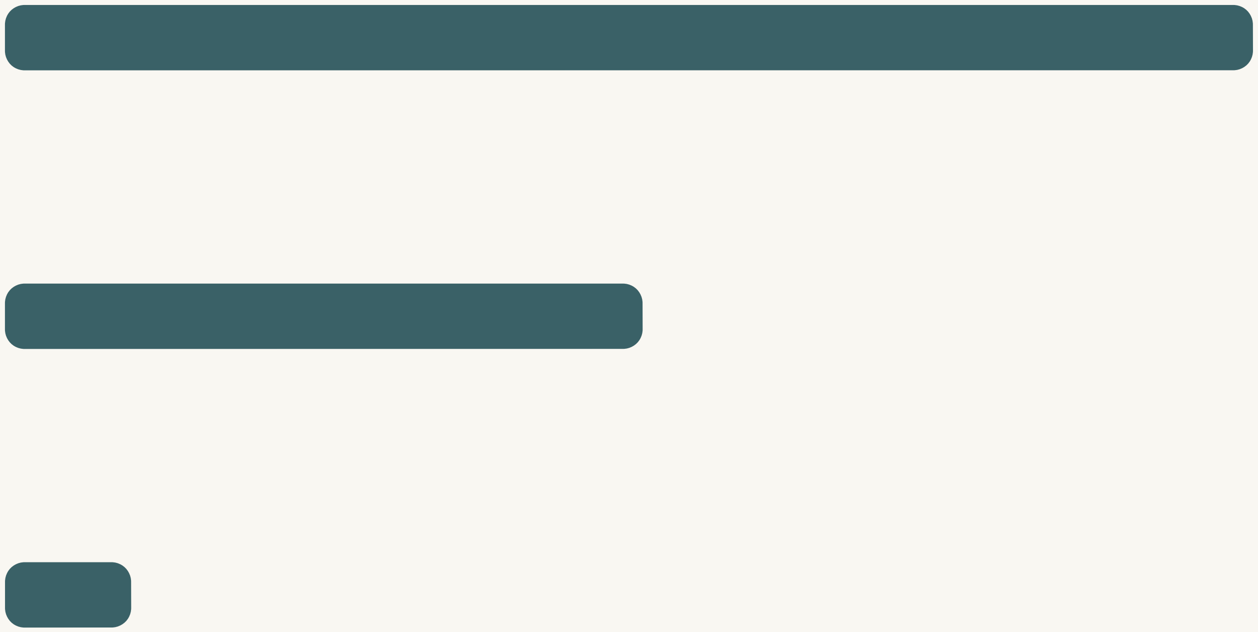
222,399 **records retrieved**
7 databases · keyword + citation

▼ de-duplication — *removes duplicate hits only*

113,643 **unique records**
after de-duplication

▼ **AI relevance screening** — on each record's title & abstract

22,484 **coded abstracts**
22,605 relevant; 121 had no abstract to code



Only ~20% of unique records passed AI relevance screening (22,605 of 113,643) — the rest were off-topic, not duplicates.

How accurate is AI screening? — from prior research

Before any coding, every record has to be judged relevant or not — the first filter, deciding which studies even enter the review. In earlier published validation, the AI was compared against the expert human teams who screened by hand.

Screening — deciding which studies belong

97.2%

of relevant studies — vs **91.6%** for humans

5 meta-analyses · 30,150 abstracts screened

What the AI is being compared to

The human baseline was thorough — every record judged by hand:

2× → **1 pass**

two reviewers + a 3rd → the AI

≈ 98–99% less time — days or months of work, done in minutes to an hour.

How the accuracy was measured

The AI and the human team each screened every title and abstract, deciding relevant or not. Where the two disagreed, the record was re-checked to establish which studies genuinely belonged. The figure shown is the percentage of those relevant studies a method correctly kept in rather than missed, averaged across the five datasets.

Mehr, A., Howard, J. L., Nouroozi, C., Khorramnazari, B., Banks, G. C., Tay, L., Cuijpers, P., Miguel, C., Harrer, M., Meyer, J. P., Stanley, D. J., Wang, X., Luo, G., Huo, B., Liu, J., & Rousseau, D. M. (2026). *Improving Evidence Synthesis with Artificial Intelligence* [Preprint].

WHAT THE CODED ABSTRACTS REVEAL

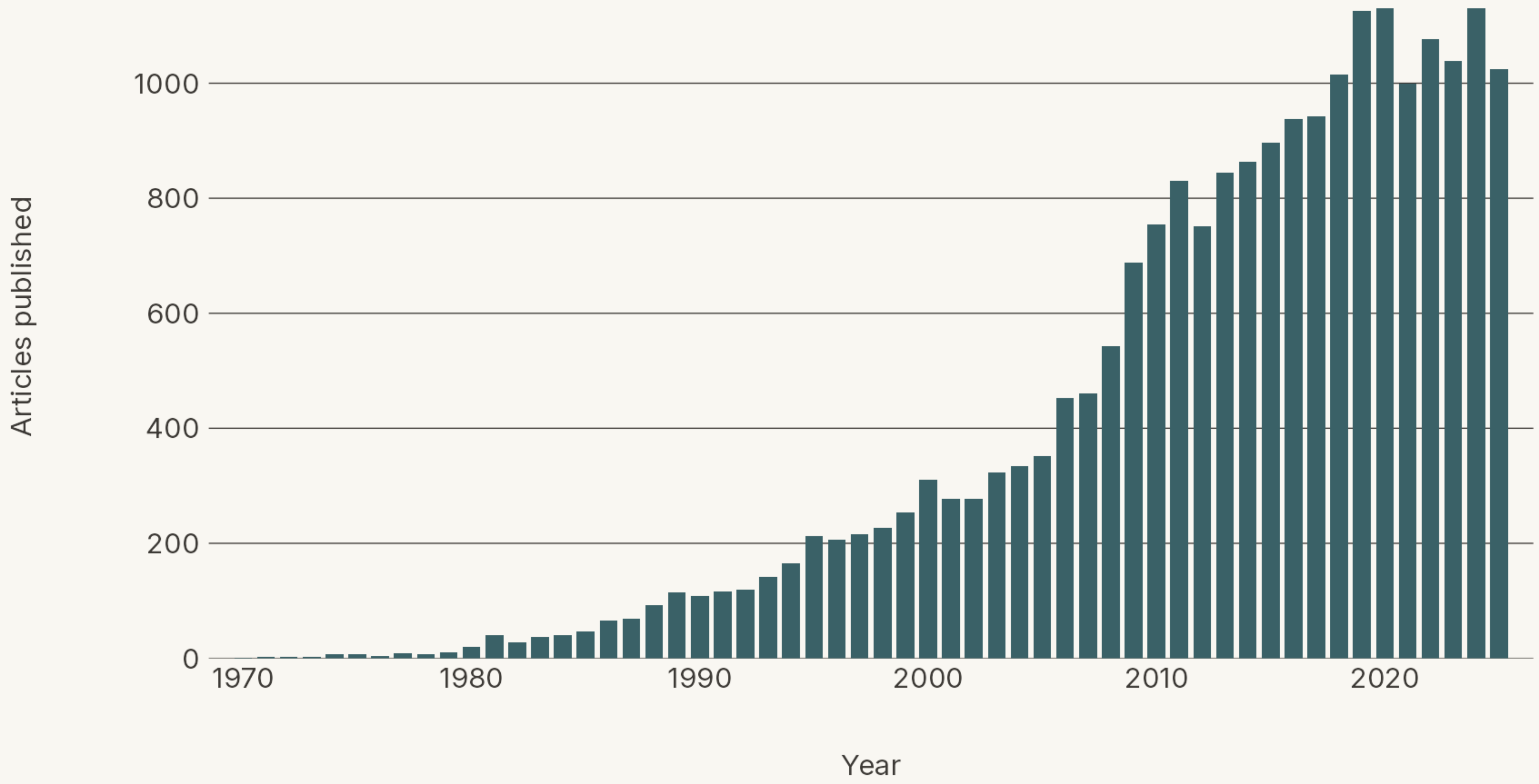
Abstract Analysis

How the literature has grown — and what it studies.

How quickly has the literature on organizational commitment grown? Here's the annual count of articles in our collection.

Articles Published Per Year

Annual count of unique abstracts, 1970–2025 (n = 21,753 of 22,484)



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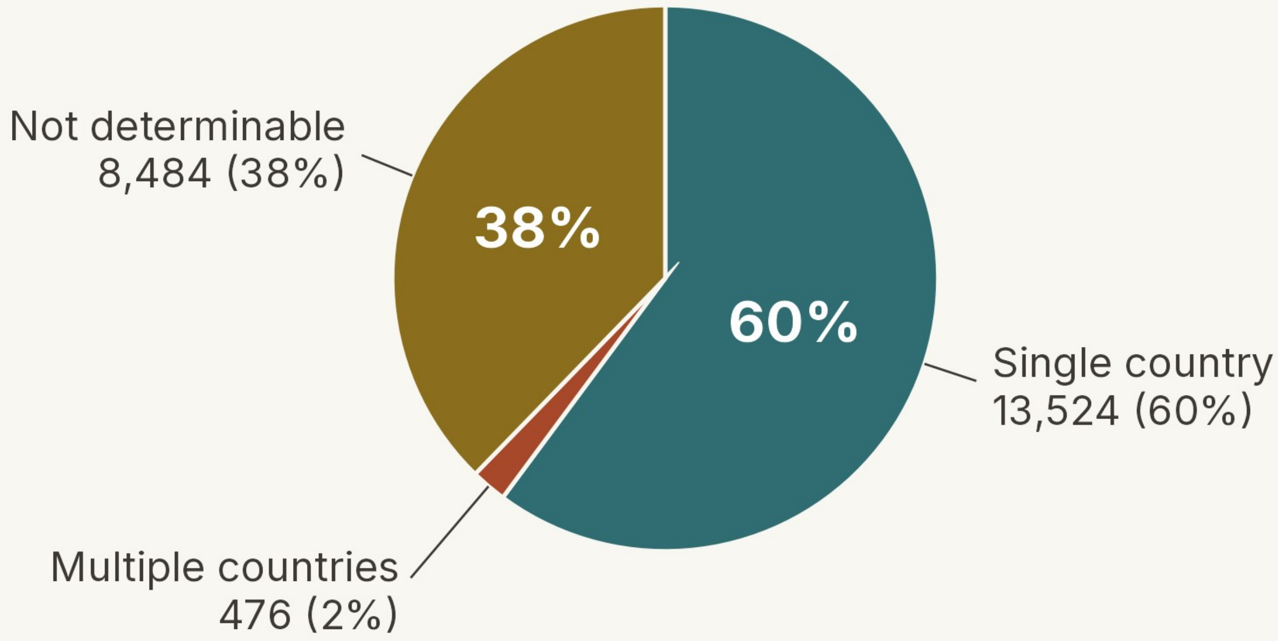
And how research on organizational commitment has spread around the world over five decades.

How many of the articles can we place geographically?

Country-coding status across all 22,484 unique abstracts



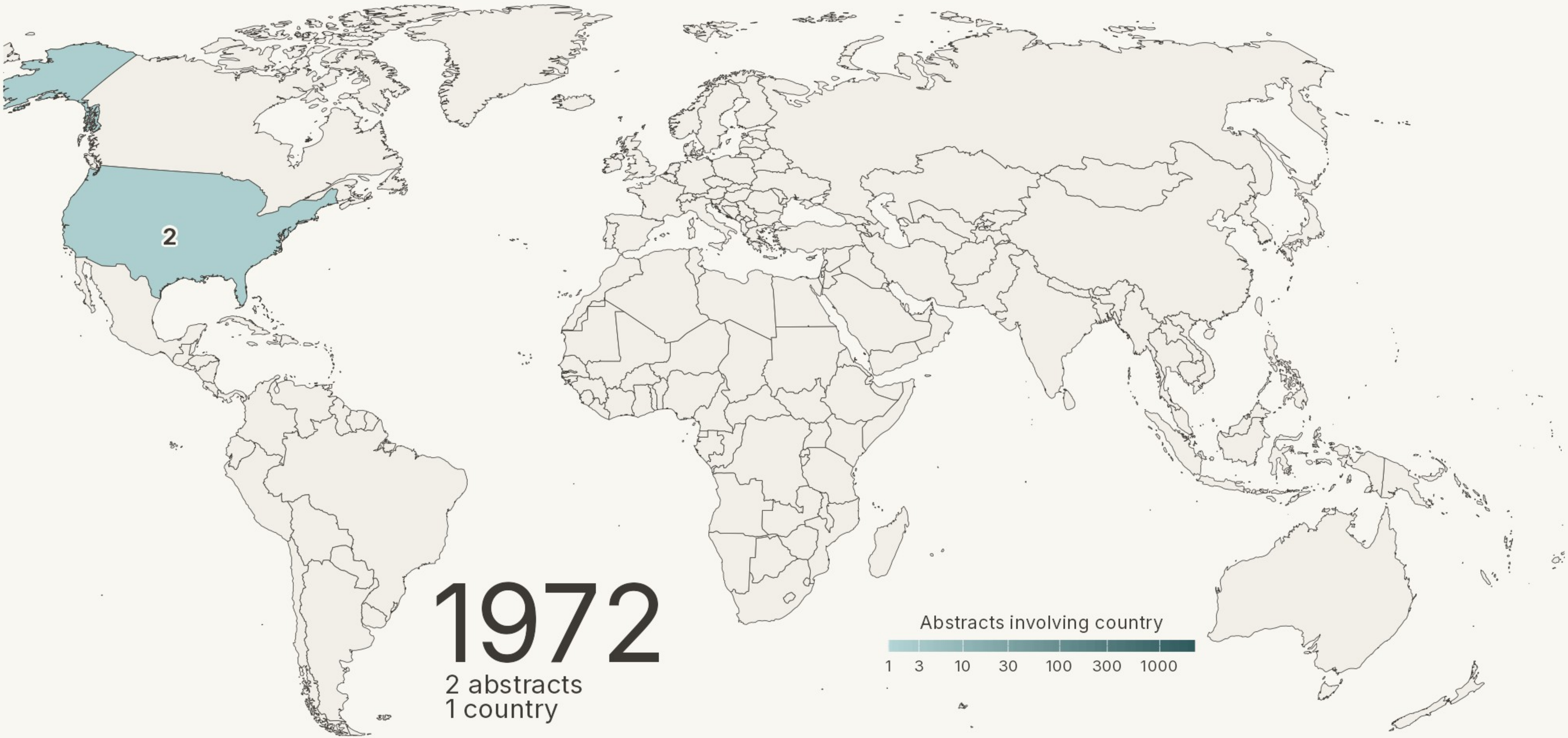
How country of origin was determined

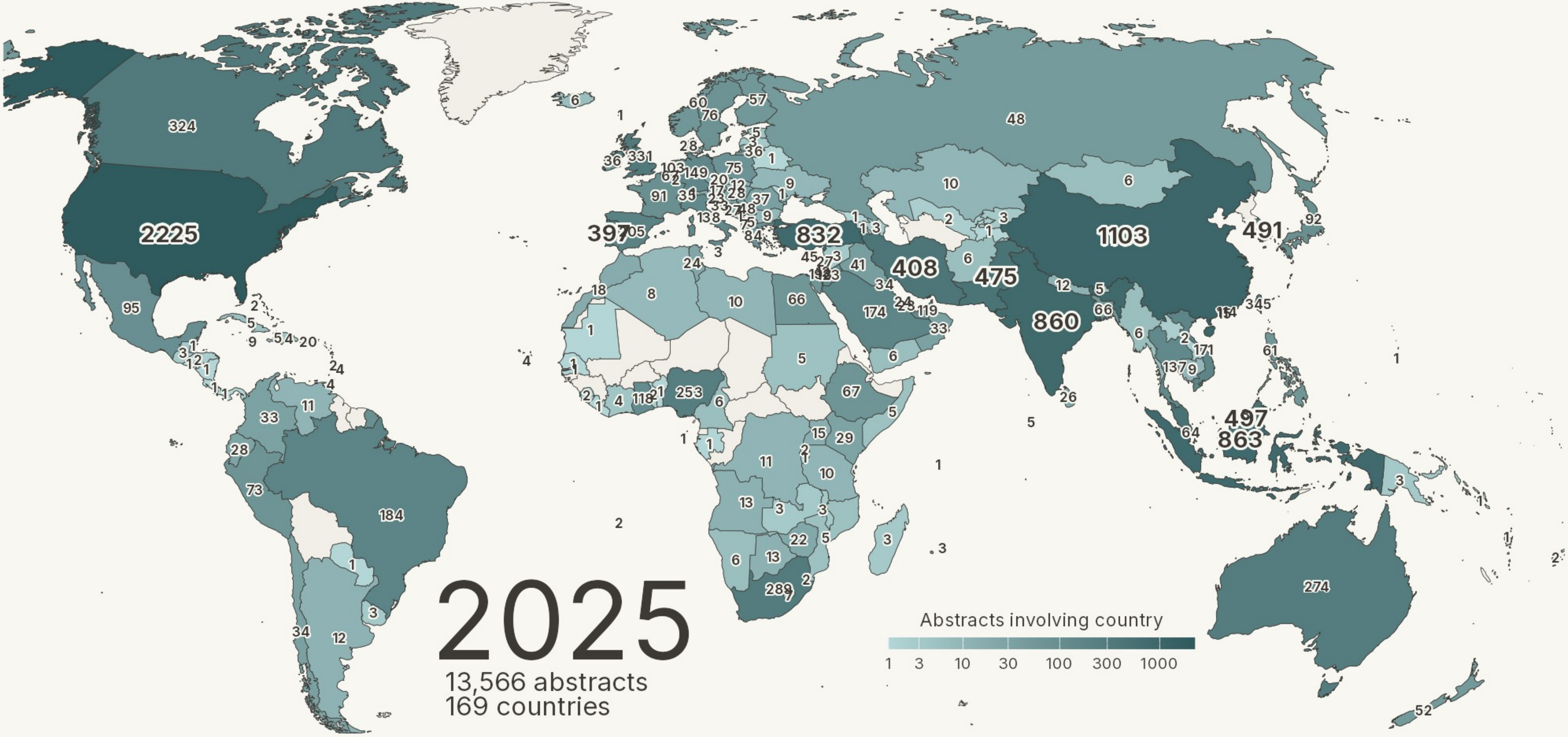
Country coding criteria (each YES / NO):

- Is a country named directly in the abstract?
- Does a stated city or region map to a country?
- Does a named company or organization map to a country?
- Do all authors' affiliations point to the same country?

Recovering undetermined cases:

- For undetermined cases, author affiliations were retrieved from OpenAlex via DOI lookup and passed to the model, resolving 14 previously undetermined items.


2
1972
2 abstracts
1 country
Abstracts involving country
1
3
10
30
100
300
1000


2025
13,566 abstracts
169 countries
Abstracts involving country
1 3 10 30 100 300 1000

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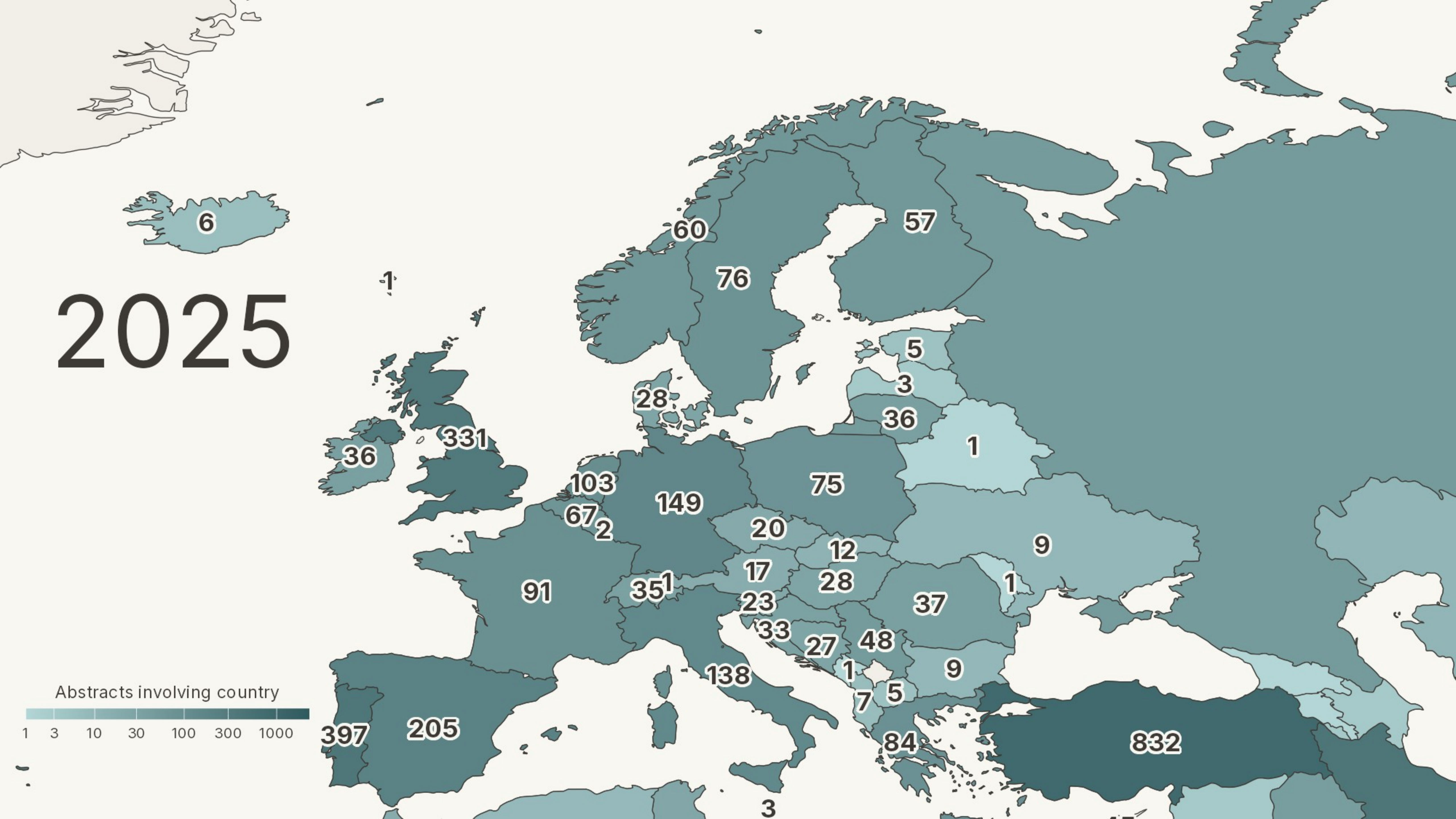
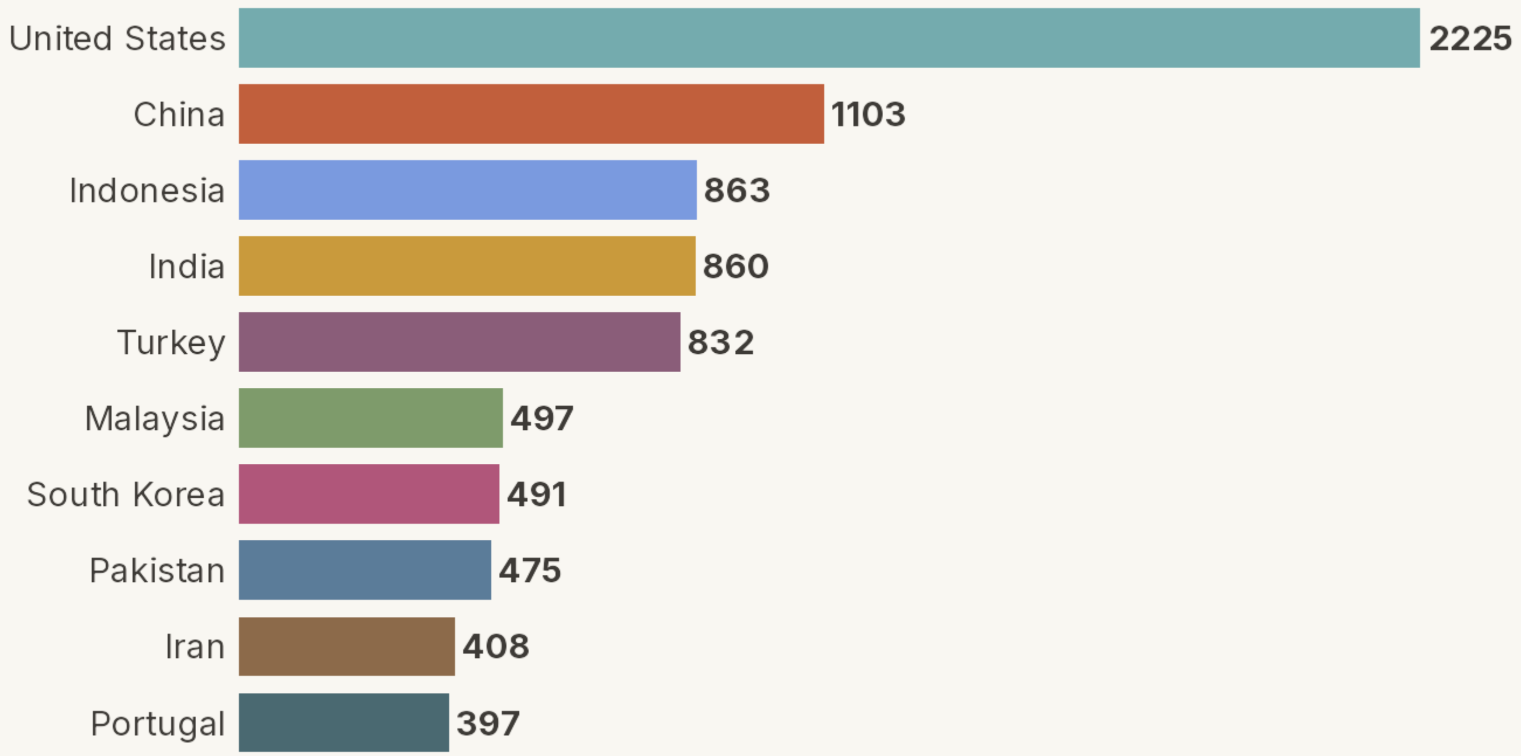

2025
Abstracts involving country
1
3
10
30
100
300
1000
6
1
60
76
57
28
5
3
36
331
36
1
103
149
75
67
2
20
9
12
17
91
35
1
28
1
23
37
33
27
48
138
1
9
7
5
397
205
84
832
3

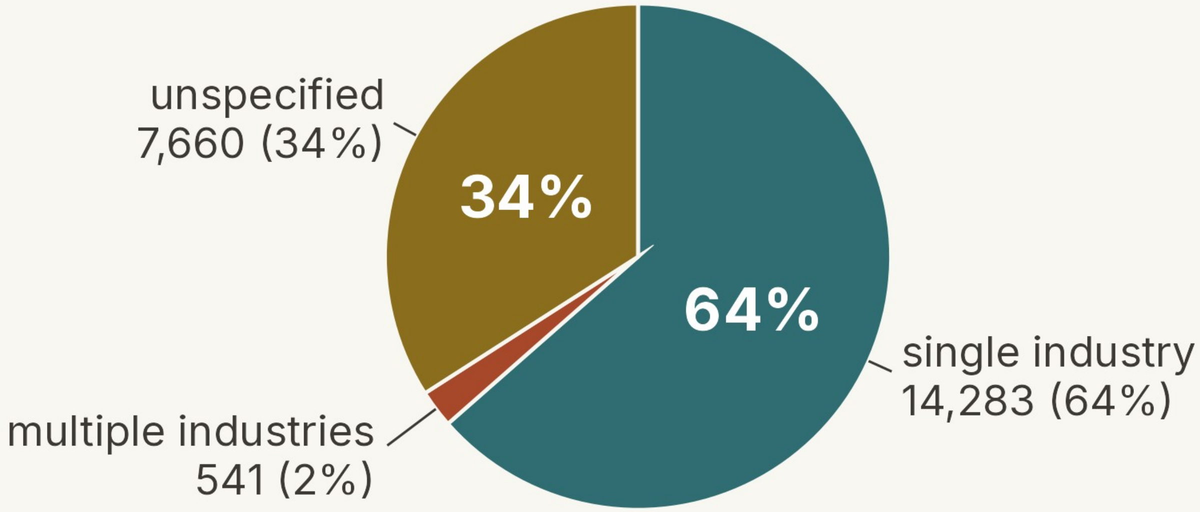
Top 10 Countries by Cumulative Samples

2025



Industry

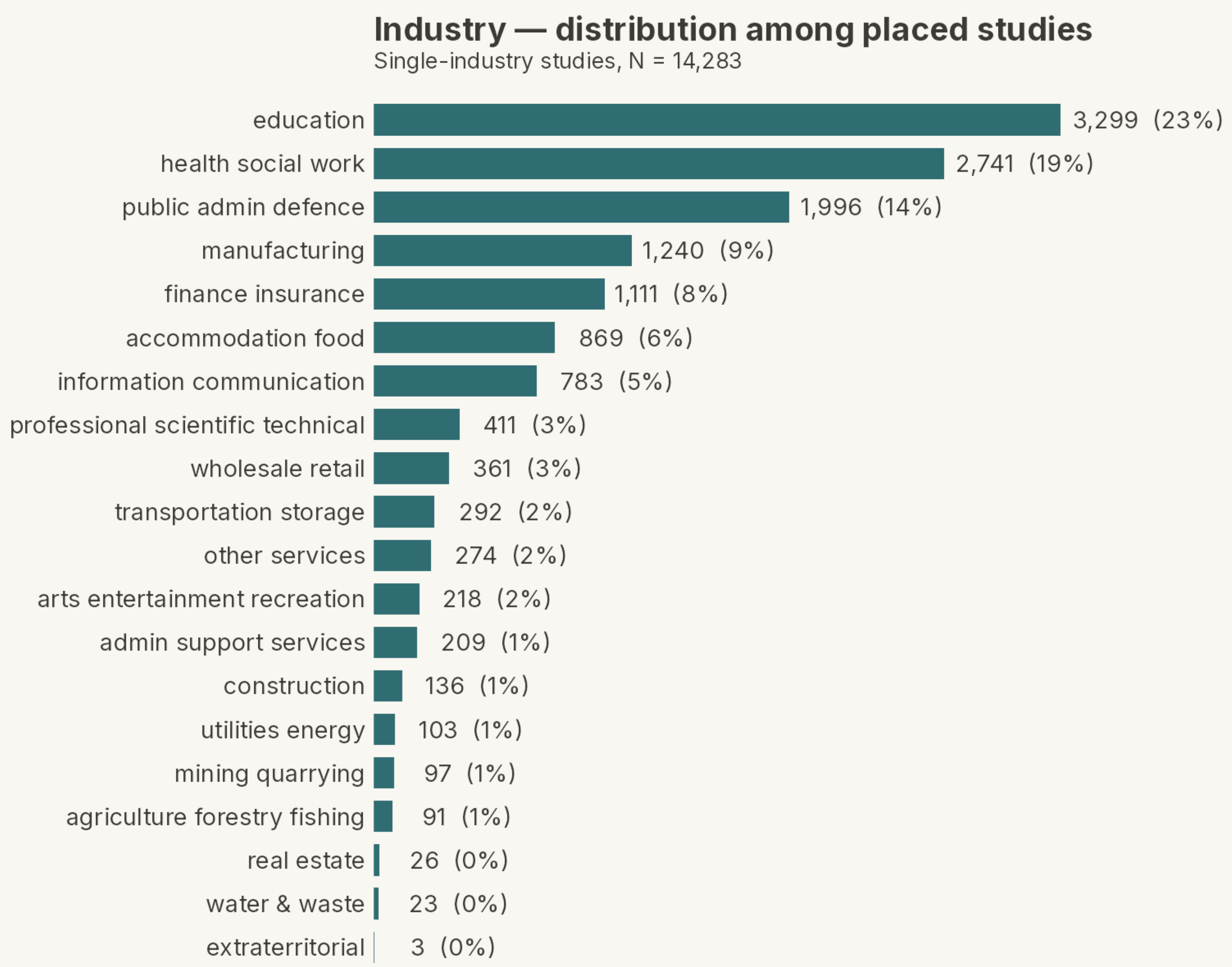
Full coded sample, N = 22,484



Industry taxonomy — ISIC Rev. 4

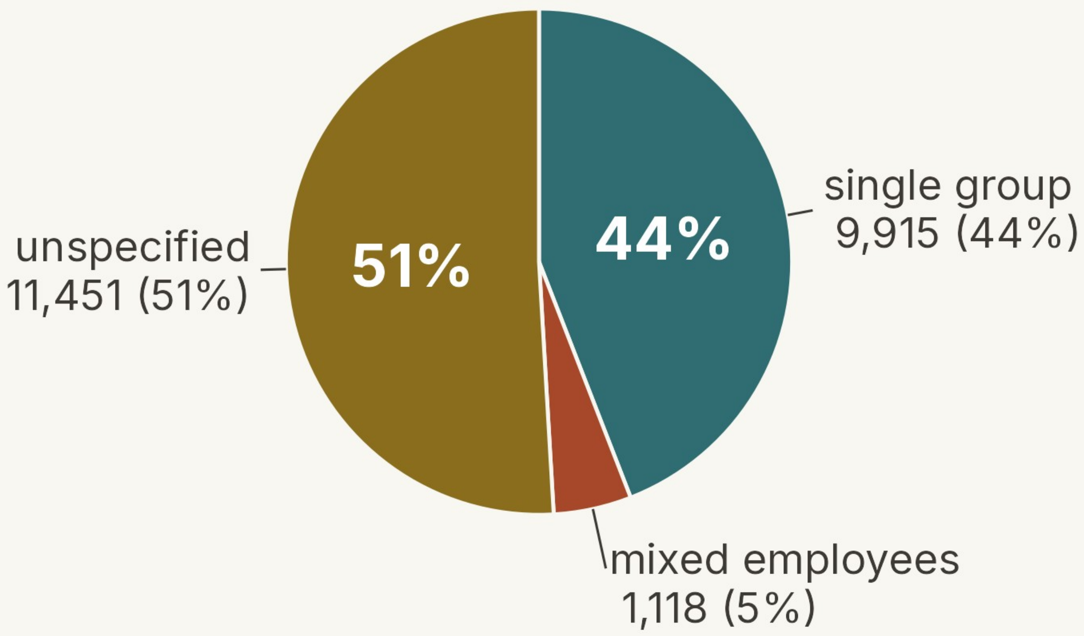
The UN's standard industry taxonomy — 21 letter-coded sections, A (agriculture, forestry & fishing) through U (extraterritorial organizations).

[Taxonomy Details Website]


Industry — distribution among placed studies
Single-industry studies, N = 14,283
education
3,299 (23%)
health social work
2,741 (19%)
public admin defence
1,996 (14%)
manufacturing
1,240 (9%)
finance insurance
1,111 (8%)
accommodation food
869 (6%)
information communication
783 (5%)
professional scientific technical
411 (3%)
wholesale retail
361 (3%)
transportation storage
292 (2%)
other services
274 (2%)
arts entertainment recreation
218 (2%)
admin support services
209 (1%)
construction
136 (1%)
utilities energy
103 (1%)
mining quarrying
97 (1%)
agriculture forestry fishing
91 (1%)
real estate
26 (0%)
water & waste
23 (0%)
extraterritorial
3 (0%)

Occupation

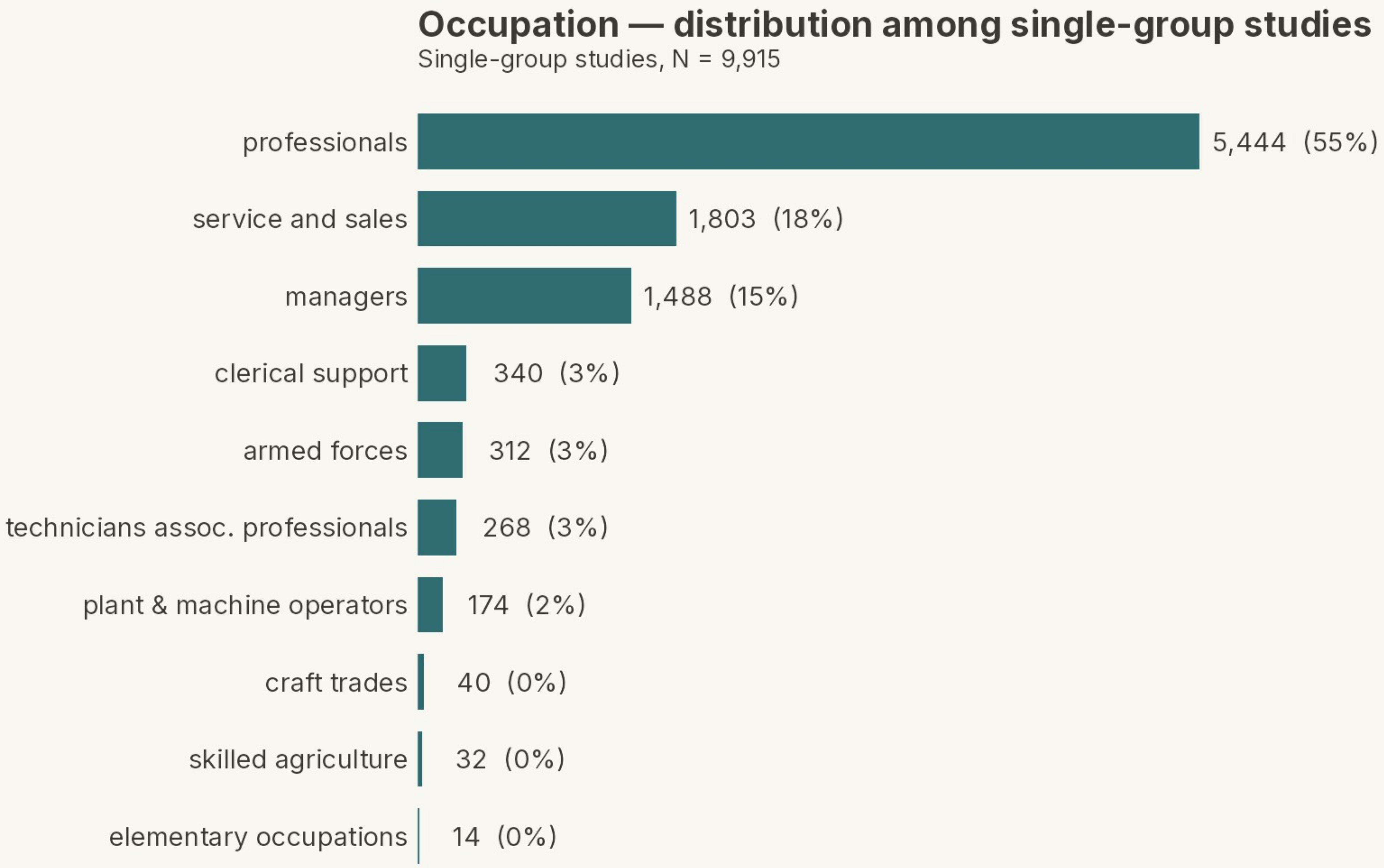
Full coded sample, N = 22,484



Occupation taxonomy — ISCO-08

The ILO's standard occupation taxonomy (the counterpart to ISIC) — 10 major groups, 0 (armed forces) through 9 (elementary occupations).

[Taxonomy Details Website]


Occupation — distribution among single-group studies
Single-group studies, N = 9,915
professionals
5,444 (55%)
service and sales
1,803 (18%)
managers
1,488 (15%)
clerical support
340 (3%)
armed forces
312 (3%)
technicians assoc. professionals
268 (3%)
plant & machine operators
174 (2%)
craft trades
40 (0%)
skilled agriculture
32 (0%)
elementary occupations
14 (0%)

Finally, how is “organizational commitment” actually used in the literature? It plays four primary roles — as a predictor, a criterion, a correlate, and a mediator — and the balance has shifted over time.

How the four roles were assigned

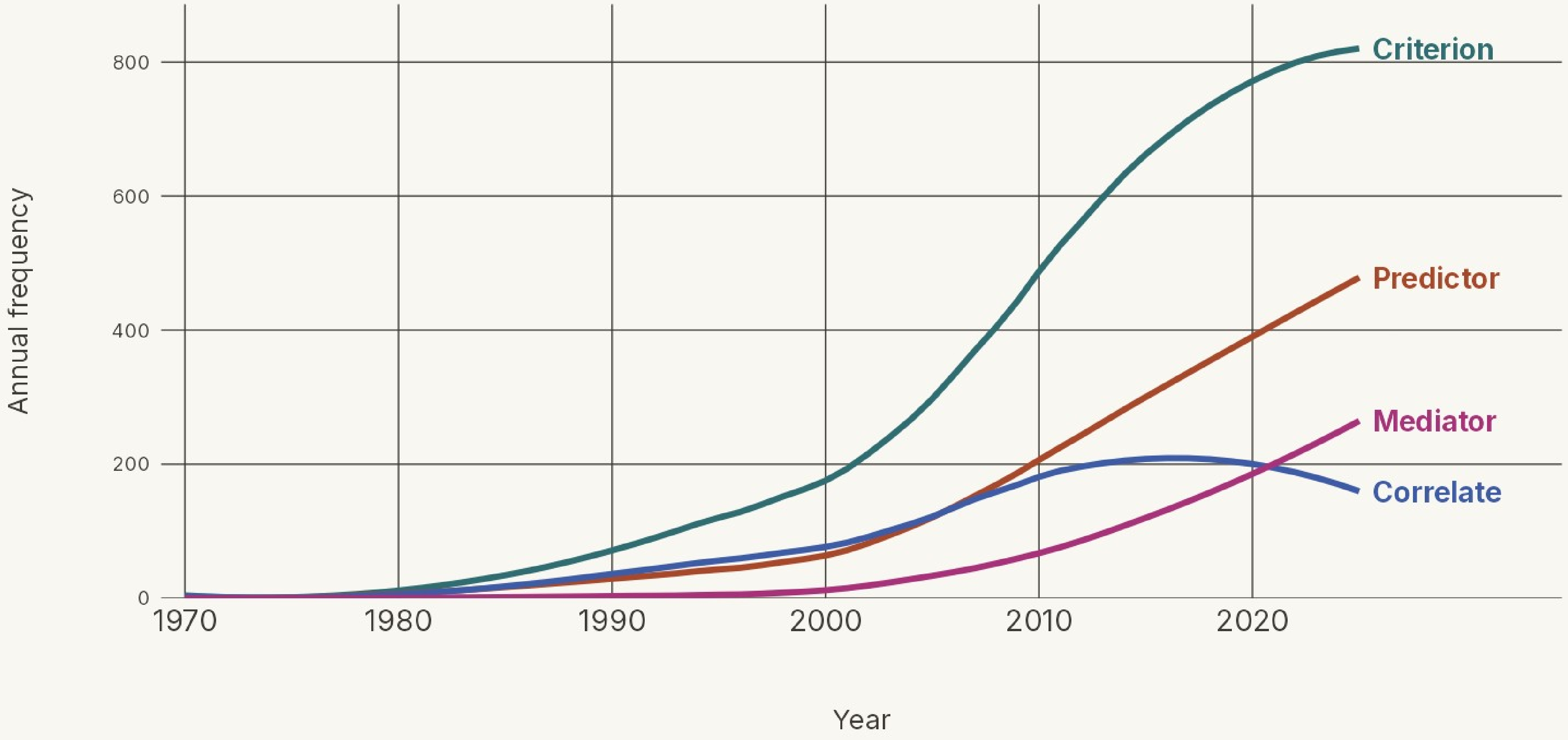
We trained the AI to build a path model from each abstract's text. A variable's position in that path model determined which of the four roles it plays.

How ‘organizational commitment’ is used

Annual count of abstracts using commitment in each role

How ‘organizational commitment’ is used

Annual count of abstracts using commitment in each role



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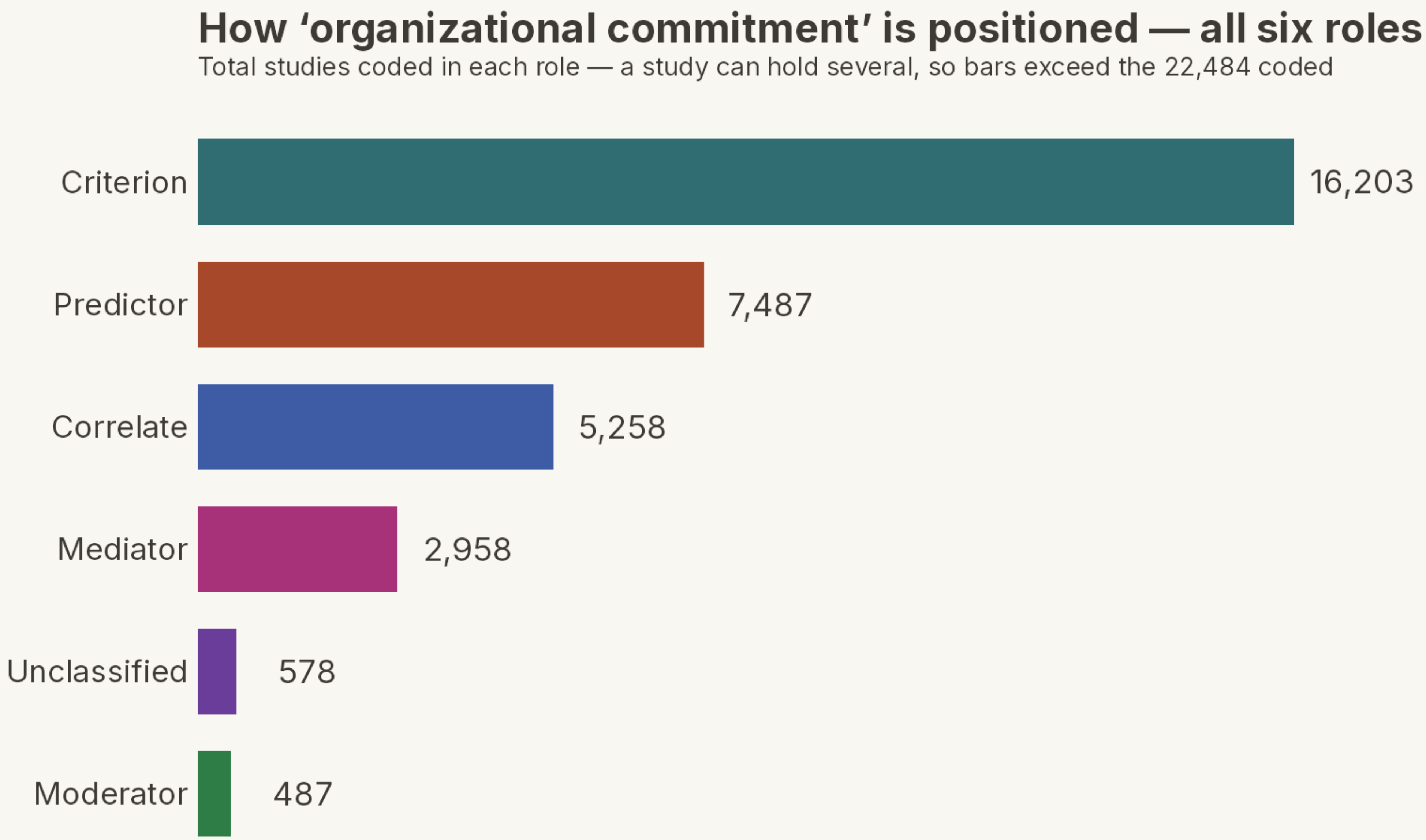


How ‘organizational commitment’ is positioned — all six roles
Total studies coded in each role — a study can hold several, so bars exceed the 22,484 coded
Criterion
16,203
Predictor
7,487
Correlate
5,258
Mediator
2,958
Unclassified
578
Moderator
487

What variables played these roles?

How did we identify and combine variable labels

STEP

1

Extract the raw labels

Direct extraction of the coded text, "as is."

STEP

2

Merge plural / spelling variants

Combine simple plural and spelling variants (e.g., "policies" → "policy").

STEP

3

Group by meaning

Group labels with the same meaning into broad families.

Steps 2 and 3 by hand would be formidable tasks — AI makes them manageable.

How many distinct variables did we extract?

After merging spelling and plural variants — before grouping by meaning

Step 2

Merge plural / spelling variants

Predictors of commitment

14,390

distinct variables

Outcomes of commitment

3,320

distinct variables

One construct, every language

For every study, the AI extracts the variables it examines — reading the abstract in its own language — and records each under one English label, so studies in any language pool into the same construct.

As written across languages

job satisfaction	English
工作满意度	Chinese
satisfação no trabalho	Portuguese
iş tatmini	Turkish
satisfacción laboral	Spanish
satisfaction au travail	French
직무 만족	Korean
الرضا الوظيفي	Arabic

The unified label

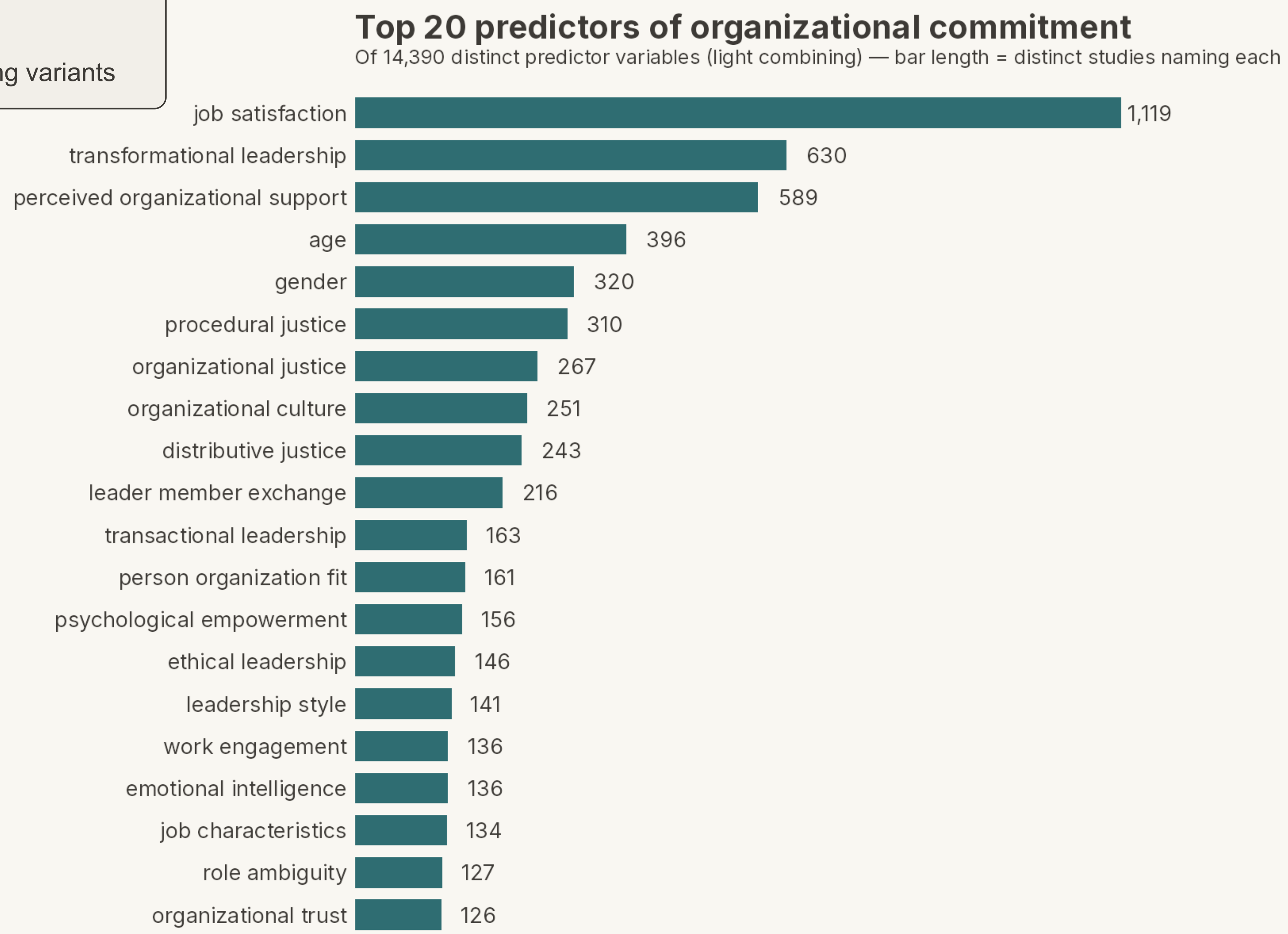
job_satisfaction

one English label · **2,085** studies pooled

The alternative — human coding — would either **miss the non-English articles** or require **hiring multiple language-proficient coders**. The AI needs neither.

Step 2

Merge plural / spelling variants

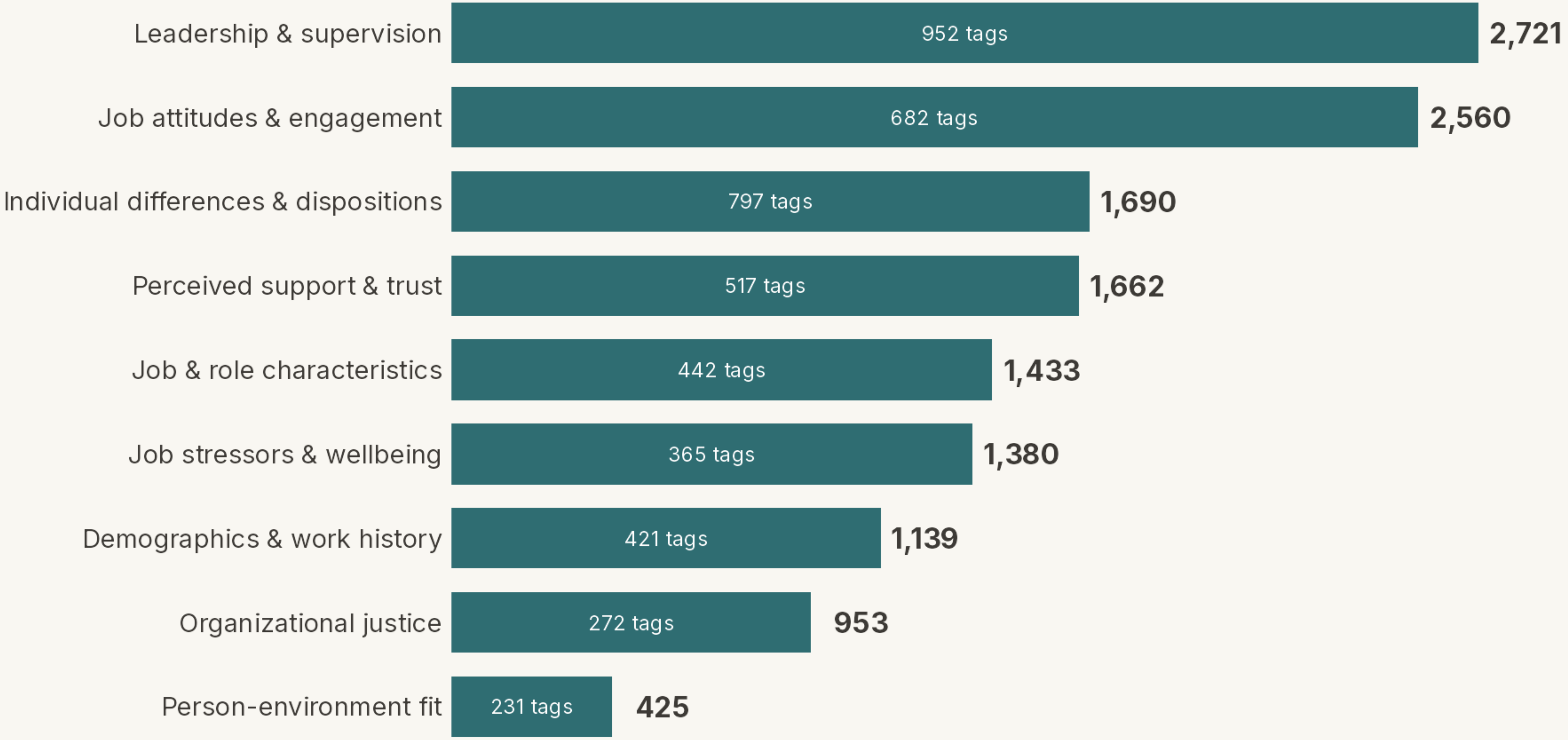


Step 3

Group by meaning

Predictors of organizational commitment, in nine categories

Bar & end label = distinct studies naming ≥1 predictor in the category.
Inside each bar = number of variables (tags) collapsed into it. (“Other” not shown.)

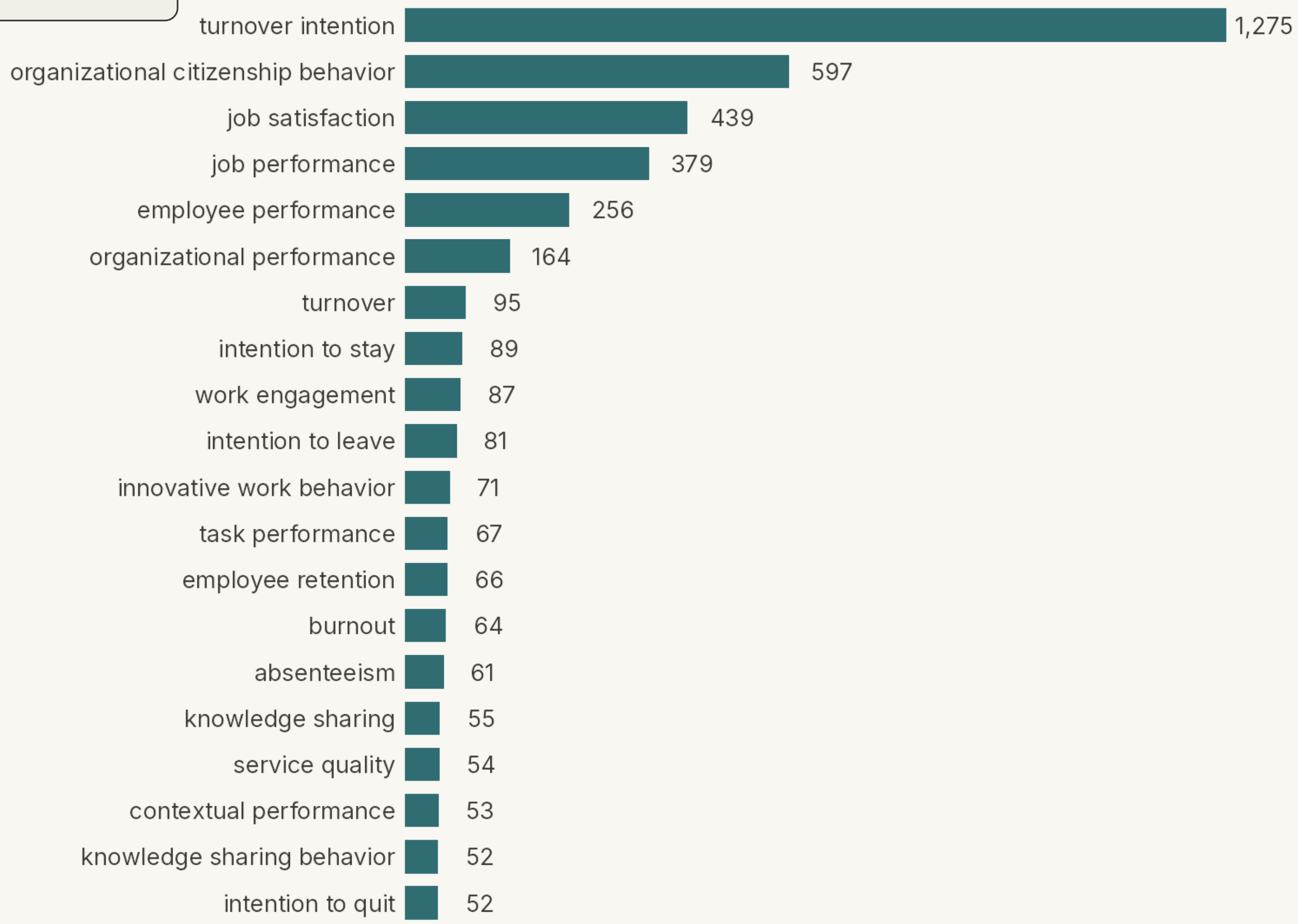


Step 2

Merge plural / spelling variants

Top 20 outcomes of organizational commitment

Of 3,320 distinct outcome variables (light combining) — bar length = distinct studies naming each

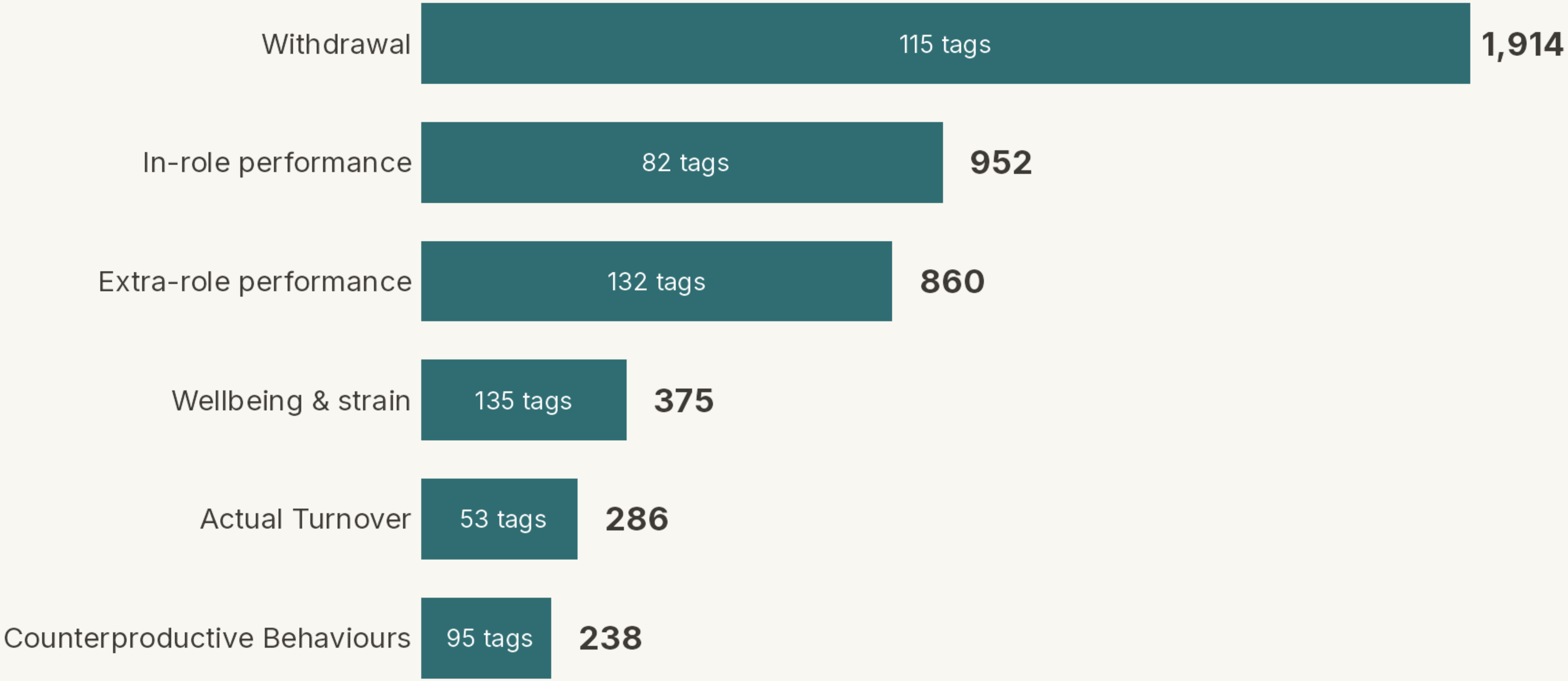


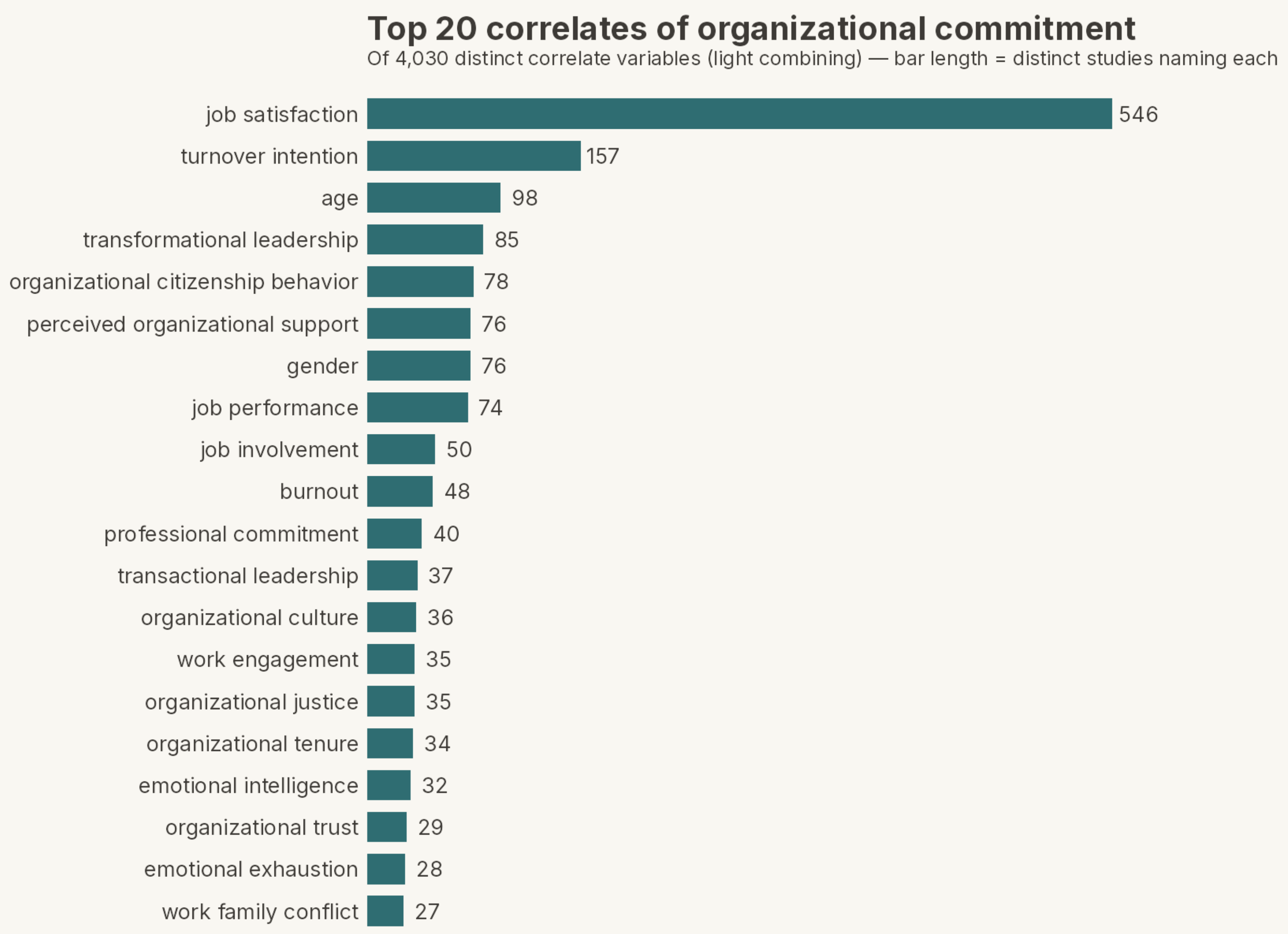
Step 3

Group by meaning

Outcomes of organizational commitment, in six categories

Bar & end label = distinct studies naming ≥1 outcome in the category.
Inside each bar = number of variables (tags) collapsed into it.




Top 20 correlates of organizational commitment
Of 4,030 distinct correlate variables (light combining) — bar length = distinct studies naming each
job satisfaction 546
turnover intention 157
age 98
transformational leadership 85
organizational citizenship behavior 78
perceived organizational support 76
gender 76
job performance 74
job involvement 50
burnout 48
professional commitment 40
transactional leadership 37
organizational culture 36
work engagement 35
organizational justice 35
organizational tenure 34
emotional intelligence 32
organizational trust 29
emotional exhaustion 28
work family conflict 27

FROM ABSTRACTS TO ANALYSIS

Next Steps

The work ahead, in four steps.

How accurate is AI coding? — from prior research

The labour-intensive part is the coding itself — reading each study and pulling out every sample size, mean, correlation, and reliability by hand. In earlier published validation, the AI was compared against the expert human teams who did exactly that.

Data extraction — the coding itself

97.0%

AI accuracy — vs **92.1%** for expert human coders

4 meta-analyses · 632 studies · 20,476 data points

What the AI is being compared to

One of the four meta-analyses, single-coded by hand:

130 hrs → **4 hrs**

one coder ≈ 32× faster the AI

12,025 numbers from 390 studies — one coder, about 20 min a study.

How the accuracy was measured

The AI and the human team each extracted every value independently. Matches were counted correct; disagreements were re-checked against the source article by an expert to decide which side was right. Each method's accuracy is the percentage of values confirmed correct, averaged across the four datasets.

Mehr, A., Howard, J. L., Nouroozi, C., Khorramnazari, B., Banks, G. C., Tay, L., Cuijpers, P., Miguel, C., Harrer, M., Meyer, J. P., Stanley, D. J., Wang, X., Luo, G., Huo, B., Liu, J., & Rousseau, D. M. (2026). *Improving Evidence Synthesis with Artificial Intelligence* [Preprint].

Next steps

From the coded abstracts to a completed meta-analysis — four steps from here.

1
Download the articles

Retrieve the full texts behind the coded abstracts.

2
Code the articles

Extract effect sizes and study details from each.

3
Formalize the variable taxonomies

Finalize the construct groupings from these results.

4
Begin analyses

Run the meta-analytic models.

SCAN TO VIEW & SHARE THESE SLIDES

tinyurl.com/icap2026-commitment

The download includes a Methodological Deep Dive we didn't have time to present.

Toward an AI-assisted Comprehensive Meta-analysis of the Organizational Commitment Literature

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UNIVERSITY OF GUELPH

the SYNTHESIS COMPANY of CALIFORNIA

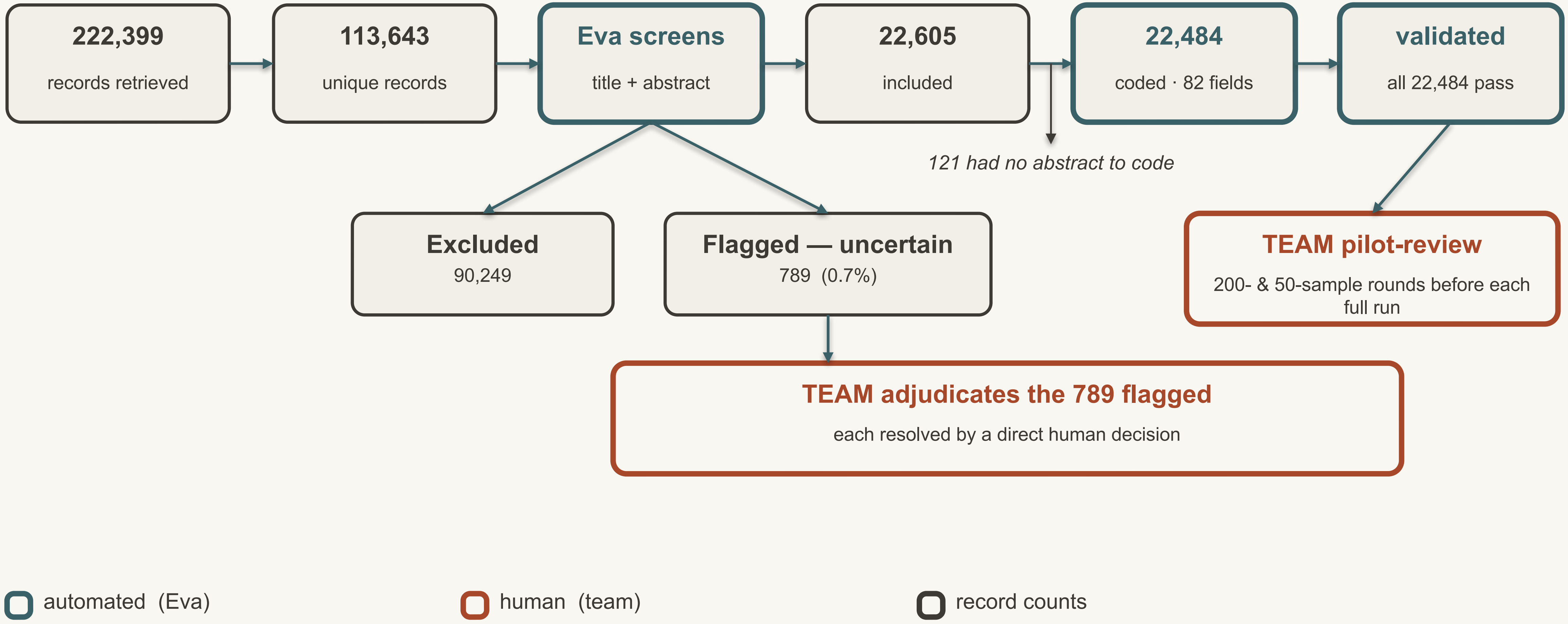
FOR READERS OF THE DOWNLOADED SLIDES

Methodological Deep Dive

The pipeline, calibration, coding scheme, and full counts — detail we skip in the talk.

The workflow, end to end

Every record's path — the fan-out at screening, and the human decisions, made explicit



The pipeline in full

Every record's path from retrieval to coded abstract — human steps marked in brown

1	**Retrieve**	222,399 records — 7 databases, keyword + cited-reference searches
2	**De-duplicate**	113,643 unique records after removing overlapping hits
3	**Screen**	Eva reads each record's title + abstract and applies 4 inclusion criteria — primary empirical, quantitative, organizational context, commitment measured: 22,605 include · 90,249 exclude · 789 flagged uncertain (0.7%)
4	**Adjudicate**	Team resolves the 789 cases the AI flagged as uncertain — each gets a direct human decision (the AI never guesses on them)
5	**Code**	22,484 abstracts (121 of the 22,605 relevant records had no abstract to code) — 82 structured fields each, with a verbatim quote per field
6	**Validate**	All 22,484 pass automated consistency checks; team pilot-reviews before each full run

Calibration & human oversight

Humans ran a blind spot-check, then concentrated effort on the cases the AI itself flagged as uncertain

STAGE	WHEN	WHAT HAPPENED
Initial alignment	Sep 2025	40 abstracts blind-reviewed — 39/40 agreement (the one mismatch was a case Eva had flagged uncertain). Adopted the rule: when in doubt, err toward inclusion.
Uncertain-case calibration	Jan–Feb 2026	195 records from ~5,400 uncertain cases, blind-reviewed by three team members. Disagreements traced to undefined rules (missing / non-English abstracts) — the reviewers disagreed too. Rules made explicit; the whole set re-screened → uncertain cases fell from ~5,400 to 789.
Uncertainty adjudication	Feb–Mar 2026	Team hand-reviewed the 789 cases the AI flagged as uncertain — each resolved by a direct human decision rather than an AI guess.
Coding pilot	May–Jun 2026	Codebook built with the team; a 50-sample review + async re-test surfaced 5 fixes (e.g., a mandatory path-diagram step, a level-of-analysis field). Only then the full 22,484 run.

Net: roughly 1% of the 113,643 records ever needed manual screening — concentrated exactly where the abstracts were genuinely ambiguous.

What was coded

82 structured fields per abstract — each carrying a verbatim quote as its evidence

PATH-MODEL FIRST

Eva drew each study's path model, located commitment within it, and only then assigned commitment's role — predictor, outcome, mediator, or moderator.

The 82 fields capture

- Commitment role (predictor / outcome / mediator / moderator)
- Commitment type (affective / continuance / normative)
- Level of analysis (individual / unit / organization)
- Country of data collection (4-level evidence hierarchy)
- Industry — ISIC Rev. 4
- Occupation — ISCO-08
- Every predictor, outcome, mediator & moderator variable
- Variable names normalized to standard English (any source language)

Evidence, not assertion — every field carries a verbatim quote from the abstract to support it.

The numbers — and where they come from

Counts are verified from this project's database; accuracy figures belong to the validation study

THIS PROJECT — DATABASE-VERIFIED

222,399	records retrieved
113,643	unique after de-duplication
22,605	included · 90,249 excluded
789	flagged uncertain — all human-adjudicated
~1%	of records needed any manual screening
22,484	abstracts coded · 82 fields each
39/40	project blind-check agreement

FROM THE VALIDATION STUDY — MEHR ET AL.

Screening

97.2% / 91.6% vs expert human screeners

Coding

97.0% / 92.1% vs expert human coders

Tested on

5 meta-analyses · 30,150 screening documents

Sensitivity and specificity are properties of the validation study, not of this meta-analysis. This project reports only its own database-verified counts and the 39/40 blind-check.

SEED STUDIES FOR THE CITED-REFERENCE SEARCH

Foundational Works

What we mean by “foundational” — the studies whose citations we followed in the cited-reference search.

Foundational works

The cited-reference search retrieved every record citing any of these 14 works

Allen, N. J., & Meyer, J. P. (1990). The measurement and antecedents of affective, continuance and normative commitment to the organization. *Journal of Occupational Psychology, 63*(1), 1–18.

Allen, N. J., & Meyer, J. P. (1996). Affective, continuance, and normative commitment to the organization: An examination of construct validity. *Journal of Vocational Behavior, 49*(3), 252–276.

Cook, J., & Wall, T. (1980). New work attitude measures of trust, organizational commitment and personal need non-fulfilment. *Journal of Occupational Psychology, 53*(1), 39–52.

Klein, H. J., Cooper, J. T., Molloy, J. C., & Swanson, J. A. (2014). The assessment of commitment: Advantages of a unidimensional, target-free approach. *Journal of Applied Psychology, 99*(2), 222–238.

Meyer, J. P., & Allen, N. J. (1984). Testing the “side-bet theory” of organizational commitment: Some methodological considerations. *Journal of Applied Psychology, 69*(3), 372–378.

Meyer, J. P., & Allen, N. J. (1991). A three-component conceptualization of organizational commitment. *Human Resource Management Review, 1*(1), 61–89.

Meyer, J. P., & Allen, N. J. (1997). *Commitment in the workplace: Theory, research, and application*. Sage Publications.

Foundational works

The cited-reference search retrieved every record citing any of these 14 works

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